Lakshyam: a story-driven life simulation for financial literacy

A browser game in which a Class 11–12 student lives one whole life, from age 11 to 36, and feels what money choices do over time.

1. Problem and solution

The brief asks for a game that teaches financial literacy to Class 11–12 students from underserved communities: simple, relatable, persona-led, ready for Indian languages, testing learning through play, intuitive, and usable by NGOs. It must cover planning and budgeting, saving and managing money, understanding investing, and staying financially safe, while educating rather than recommending products and treating simulations as assumptions, not promises.

Our reading: these students do not mainly lack information; they lack a safe place to make money mistakes before the mistakes are real. Real consequences arrive years later, so the link between a choice and its cost is never felt. **Lakshyam** (“goal”) gives them that place. The player picks a look, a language and a dream, then grows up in a Telangana-style pixel village with a tenant-farming family, a grandmother mentor (Ammamma) and a best friend who always takes the shortcut. Six story chapters cover a whole life; a full life takes about 15–20 minutes and runs offline in a laptop browser.

2. Approach

- **Three parts, one contract.** A pure, seeded JavaScript **engine** (no React) holds all rules and money maths; **content** (stories, lessons, glossary, FAQ, documents) is JSON in English, Hindi and Telugu; the React **UI** only renders state and sends actions through one adapter. A shared written spec let three parallel workstreams build at the same time.
- **Curriculum first.** A fixed curriculum of 24 lessons (L01–L24) spans the four areas, from needs vs wants in childhood to insurance, investing and pensions in adulthood. Every story choice is tagged with the lesson it teaches.
- **Seeded and reproducible.** The same seed and choices give the same life. This makes *Parallel Life* fair (only the changed choice differs) and lets a teacher replay a class demo (`?seed=demo`).
- **One source of truth for numbers.** Every rate, premium, wage and cost is in one assumptions file; outcomes always show the numbers and a “Why?” explanation. No real brands or products are named.
- **Safe by design.** Illegal paths (money mule, Ponzi recruiting, loan apps) show consequences, never methods, and always offer a way back (e.g. the 1930 cyber-crime helpline, reporting in the golden hour).
- **Iterated on evidence.** At M1 the game was a card per year. Our own playthroughs found it felt like a quiz, so the main path became a branching story mode; the year-card mode is kept and still covers all 24 lessons. Simulation runs (Section 4) then drove balance fixes.

3. Features

Feature	What the player does	Teaches
Six branching chapters	Prologue; *The OTP Heist* (school); *The Telegram Guru* (college); *The Loan App* (first job); *Wedding Season* (family); *Golden Harvest* (dream). Choices open or lock later ones; each chapter ends in one of several outcomes (from “The Scam Buster” to “Bankrupt at 24”) and a flowchart shows paths not taken.	Cause and effect across a life
Save first + jars	Each chapter starts with one question: save little, some or lots. A live forecast shows when the dream arrives, how many surprises the safety jar covers and how much fun money is left. Surprises (fever, broken phone, fees, weddings) are paid from the safety jar, or borrowed at interest.	Budgeting, emergency fund, cost of borrowing
STOP warnings and timers	Before a very bad step (sharing an OTP, a loan app, renting a bank account, a “guaranteed double”) the screen flashes	Slowing down under pressure

	STOP. Some scenes run on a countdown, like a real scam call or the golden hour to report fraud.	
Evidence file	Seeing through scams (recording the call, checking SEBI registration) collects evidence that decides the final chapter.	Verifying before trusting
Choices you play	*Rat Guard* (tin box vs bank), *Interest Race* (P×R×T/100 under time), *Fine Print Hunter* (hidden loan charges; ₹19,000 in hand, ₹24,800 back), *SIP Garden* (units, NAV, diversification, crash year), *Flood Rescue* (crop insurance and documents).	Formulas and reflexes, not just reading
Phone	Messages, a bank, an invest app with fake funds, and a practice stock exchange with a live order book and seven missions (spread, slippage, news).	Investing without recommending products
Arcade and scam ads	Mini-games with occasional fake ads that look like the game's own posters (spin-to-win, ₹1 PRO AutoPay, fake bank login); clicking opens a realistic fake site, then a reveal of the red flags.	Spotting online fraud
Learning support	Gyaan lesson slides (big idea, example, myth card, calculator, quiz), a glossary, an FAQ and *Ask Ammamma*, an offline mentor answering from the FAQ.	Works in classrooms with no internet
End of life	Story ending, money ending, a per-lesson **Gyaan report** (seen / applied), **Parallel Life** (change one decision, replay the same seed) and an ending gallery.	Reflection; teacher/NGO measure
Languages and access	English, हिन्दी and తెలుగు, switchable at any time; keyboard play (arrow keys/WASD, 1–4), quiet synthesized sound, progress saved only in the browser, no accounts or personal data.	Indian-language readiness

4. Results

Automated testing. All checks pass on the final commit:

Layer	What runs	Result
Engine unit tests	121 Node tests: seeded determinism, budget and EMI rules, money conservation, lessons inside their age windows, replay / Parallel Life, endings, story-mode timers, locks and flowcharts, mini-game logic (arcade, SIP Garden, Flood Rescue, order book, scam ads).	121 / 121 pass
Simulated lives	150 random story-mode lives all reach age 36 with an ending, and every story node is reachable; 300 random chapter-mode lives: no NaN, money conserved, every lesson inside its window; careful players catch the scammer and reach the dream, reckless ones do not.	All pass
Content validator	Schema, cross-references, lesson ids and age windows, and completeness of every string in all three languages.	0 errors, 0 warnings
Browser (Playwright)	Start → language → village → dream → prologue → outcome and flowchart; STOP warning and timed OTP scene; arrow-key walking; language switching without overflow; save and continue; Ask Ammamma; layouts at 1024×768, 1280×720 and 1366×768.	Pass
Build	Production Vite build; runs fully offline as a static site.	Pass

What simulation taught us. At M2, a 900-life headless run (good, random and bad policies, 300 each) showed 0 crashes, 0 stuck lives and exact money conservation, and that choice quality maps cleanly onto outcomes: lessons applied fell from 17 (good) to 8 (random) to 0 (bad). It also exposed two real problems we would not have found by hand: careful players ended with a smaller money result than reckless ones, and some story paths skipped a few lessons. After M2 we added the *Save first* jars with a dream forecast and surprises paid from the safety jar (so preparation is visible in money, not only in lessons), and the end-of-life Gyaan report. Internal playtests also led to a worked example before the Interest Race and a clearer font for money amounts.

Limitations and next steps. We have not yet run the pre/post quiz with real students; that is the planned learning measure, alongside native-speaker review of Hindi and Telugu and a second (city) storyline. All numbers are illustrative teaching assumptions, not financial advice.